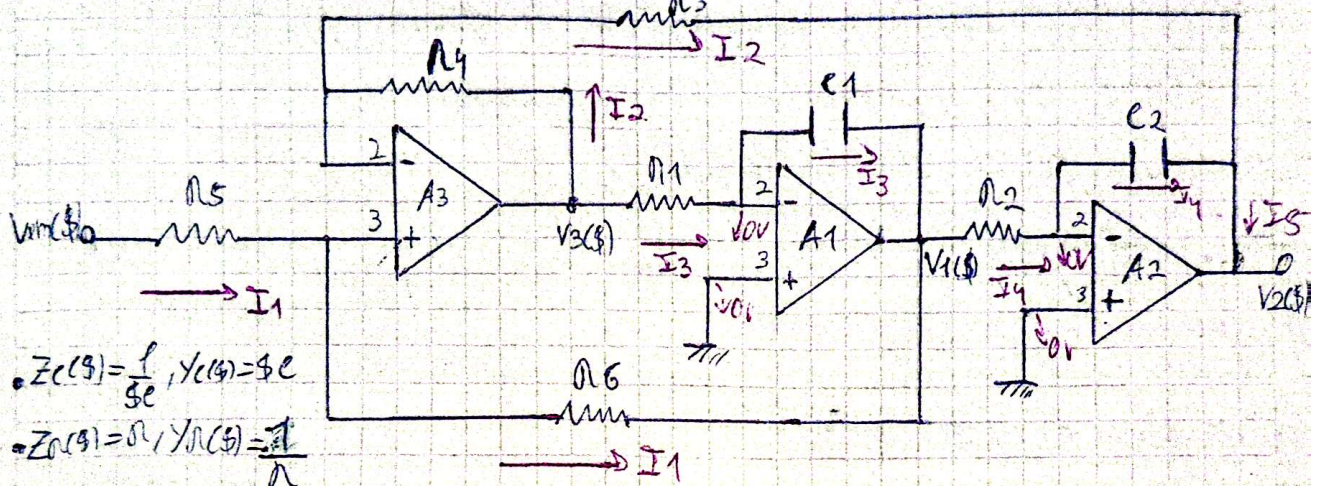


Ejercicio 5)

• Filtro de variables de estado, busquemos los transformadores

a) $\frac{V_2(s)}{V_{in}(s)}$, b) $\frac{V_3(s)}{V_{in}(s)}$, c) $\frac{V_1(s)}{V_{in}(s)}$



• Suponga amplificadores ideales $\Rightarrow$ Para A_3, A_1 y $A_2 \Rightarrow V^+ = V^-$

$$(V_{in}(s) - V_{A3}^+)Y_{R5}(s) = I_1(s) = (V_{A3}^+ - V_1(s))Y_{R6}(s)$$

$$\Rightarrow V_{in}(s) \cdot Y_{R5}(s) - V_{A3}^+ \cdot Y_{R5}(s) = V_{A3}^+ \cdot Y_{R6}(s) - V_1(s) \cdot Y_{R6}(s)$$

$$\Rightarrow V_1(s) \cdot Y_{R6}(s) = V_{A3}^+ \cdot Y_{R6}(s) + V_{A3}^+ \cdot Y_{R3}(s) - V_{in}(s) \cdot Y_{R5}(s)$$

$$V_1(s) = V_{A3}^+ \cdot \left(\frac{Y_{R6}(s) + Y_{R3}(s)}{Y_{R6}(s)} \right) - V_{in}(s) \cdot \frac{Y_{R5}(s)}{Y_{R6}(s)}$$

$$V_1(s) = V_{A3}^+ \left(1 + \frac{Y_{R5}(s)}{Y_{R6}(s)} \right) - V_{in}(s) \cdot \frac{Y_{R5}(s)}{Y_{R6}(s)} \quad (A)$$

$$V_{A3}^+ = V_{A3}^- \Rightarrow I_2(s) = (V_3(s) - V_{A3}^-) \cdot Y_{R4}(s) = (V_{A3}^- - V_2(s)) \cdot Y_{R3}(s)$$

$$\Rightarrow V_3(s) \cdot Y_{R4}(s) - V_{A3}^+ \cdot Y_{R4}(s) = V_{A3}^+ \cdot Y_{R3}(s) - V_2(s) \cdot Y_{R3}(s)$$

$$V_3(s) \cdot Y_{R4}(s) + V_2(s) \cdot Y_{R3}(s) = V_{A3}^+ \cdot Y_{R3}(s) + V_{A3}^+ \cdot Y_{R4}(s)$$

$$V_3(s) \cdot Y_{R4}(s) + V_2(s) \cdot Y_{R3}(s) = V_{A3}^+ (Y_{R3}(s) + Y_{R4}(s)) \quad (B)$$

$$\cdot I_3(s) = (V_3(s) - V_{A1}^-) \cdot Y_{n1}(s) = (V_{A1}^- - V_1(s)) \cdot Y_{c1}(s), \quad V_{A1}^- = V_{A1}^+ = 0V$$

$$\Rightarrow (V_3(s) - 0V) Y_{n1}(s) = (0V - V_1(s)) \cdot Y_{c1}(s)$$

$$\Rightarrow V_3(s) \cdot Y_{n1}(s) = -V_1(s) \cdot Y_{c1}(s)$$

$$\boxed{V_3(s) = -V_1(s) \cdot \frac{Y_{c1}(s)}{Y_{n1}(s)}} \quad (C)$$

$$\cdot I_4(s) = (V_1(s) - V_{A2}^-) \cdot Y_{n2}(s) = (V_{A2}^- - V_2(s)) \cdot Y_{c2}(s), \quad V_{A2}^- = V_{A2}^+ = 0V$$

$$\Rightarrow (V_1(s) - 0V) Y_{n2}(s) = (0V - V_2(s)) \cdot Y_{c2}(s)$$

$$\Rightarrow V_1(s) \cdot Y_{n2}(s) = -V_2(s) \cdot Y_{c2}(s)$$

$$\Rightarrow \boxed{-V_1(s) \cdot \frac{Y_{n2}(s)}{Y_{c2}(s)} = V_2(s)} \quad (D)$$

Reemplazo (C) y (D) en (B)

$$-V_1(s) \cdot \frac{Y_{c1}(s)}{Y_{n1}(s)} \cdot Y_{n4}(s) - V_1(s) \cdot \frac{Y_{c2}(s)}{Y_{c2}(s)} \cdot Y_{n3}(s) = V_{A3}^+ (Y_{n3}(s) + Y_{n4}(s))$$

$$\frac{-V_1(s) \cdot \frac{Y_{c1}(s) \cdot Y_{n4}(s)}{Y_{n1}(s)}}{Y_{n3}(s) + Y_{n4}(s)} - \frac{V_1(s) \cdot \frac{Y_{n2}(s) \cdot Y_{n3}(s)}{Y_{c2}(s)}}{Y_{n3}(s) + Y_{n4}(s)} = V_{A3}^+$$

$$\boxed{-V_1(s) \frac{Y_{c1}(s) \cdot Y_{n4}(s)}{Y_{n1}(s) (Y_{n3}(s) + Y_{n4}(s))} - V_1(s) \frac{Y_{n2}(s) \cdot Y_{n3}(s)}{Y_{c2}(s) (Y_{n3}(s) + Y_{n4}(s))} = V_{A3}^+} \quad (E)$$

Reemplazo (E) en (A)

$$V_1(s) = \left[-V_{A3}^+ \frac{Y_{n1}(s) (Y_{n3}(s) + Y_{n4}(s))}{Y_{c1}(s) \cdot Y_{n4}(s)} - V_{A3}^+ \frac{Y_{c2}(s) (Y_{n3}(s) + Y_{n4}(s))}{Y_{n2}(s) \cdot Y_{n3}(s)} \right] \cdot \left(1 + \frac{Y_{n5}(s)}{Y_{n6}(s)} \right) - \frac{V_{n5}(s) \cdot Y_{n5}(s)}{Y_{n6}(s)}$$

$$\cdot \text{Área (C)} \Rightarrow a = \frac{Y_{c1}(s) \cdot Y_{n4}(s)}{Y_{n1}(s) (Y_{n3}(s) + Y_{n4}(s))}$$

$$\cdot \text{Área (D)} \Rightarrow b = \frac{Y_{n2}(s) \cdot Y_{n3}(s)}{Y_{c2}(s) (Y_{n3}(s) + Y_{n4}(s))}$$

$$\cdot \text{Área (E)} \Rightarrow c = 1 + \frac{Y_{n5}(s)}{Y_{n6}(s)}, \quad \cdot \text{Área (F)} \Rightarrow d = \frac{Y_{n5}(s)}{Y_{n6}(s)}$$

$$\Rightarrow V_1(\$) = -V_1(\$) \cdot a \cdot c - V_1(\$) \cdot b \cdot c - V_{im}(\$) \cdot d$$

$$\Rightarrow V_1(\$) + V_1(\$) \cdot a \cdot c + V_1(\$) \cdot b \cdot c = -V_{im}(\$) \cdot d$$

$$\Rightarrow V_1(\$) [1 + a \cdot c + b \cdot c] = -V_{im}(\$) \cdot d$$

$$\Rightarrow \boxed{\frac{V_1(\$)}{V_{im}(\$)} = - \frac{d}{1 + a \cdot c + b \cdot c}} \quad (F)$$

$$\bullet \text{ Además } \Rightarrow \boxed{V_2(\$) = V_{im}(\$) \cdot \frac{d}{1 + a \cdot c + b \cdot c} \cdot \frac{V_{i2}(\$)}{V_{e2}(\$)}} \quad (E)$$

$$\bullet \text{ Además } \Rightarrow \boxed{V_3(\$) = V_{im}(\$) \cdot \frac{d}{1 + a \cdot c + b \cdot c} \cdot \frac{V_{e1}(\$)}{V_{i1}(\$)}} \quad (H)$$

• Para trabajar (F), (E) y (H) hay que tener trabajados la expresión: $-\frac{d}{1 + a \cdot c + b \cdot c}$

$$a = \frac{V_{e1}(\$) \cdot V_{i4}(\$)}{V_{i1}(\$) (V_{i3}(\$) + V_{i4}(\$))} = \frac{\$C_1 \cdot \frac{1}{n_4}}{\frac{1}{n_1} \left(\frac{1}{n_3} + \frac{1}{n_4} \right)} \cdot \frac{n_4}{n_4} = \frac{\$C_1}{\frac{1}{n_1} \left(\frac{1}{n_3} + \frac{1}{n_4} \right)}$$

$$a = \frac{\$C_1}{\frac{n_4}{n_1 n_3} + \frac{1}{n_1}} = \frac{\$C_1}{\frac{n_4 + n_3}{n_1 n_3}} = \boxed{\frac{\$C_1 \cdot n_1 n_3}{n_4 + n_3}} \quad (I)$$

$$b = \frac{V_{i2}(\$) \cdot V_{i3}(\$)}{V_{i2}(\$) (V_{i3}(\$) + V_{i4}(\$))} = \frac{\frac{1}{n_2} \cdot \frac{1}{n_3}}{\$C_2 \left(\frac{1}{n_3} + \frac{1}{n_4} \right)} = \frac{1}{\$C_2 n_2 n_3 \left(\frac{1}{n_3} + \frac{1}{n_4} \right)} \quad (J)$$

$$b = \frac{1}{\$C_2 n_2 n_3 \left(\frac{n_4 + n_3}{n_3 n_4} \right)} = \frac{1}{\$C_2 n_2 \left(\frac{n_4 + n_3}{n_4} \right)} = \boxed{\frac{n_4}{\$C_2 n_2 (n_4 + n_3)}} \quad (K)$$

$$c = 1 + \frac{Y_{n5}(\$)}{Y_{n6}(\$)} = 1 + \frac{1/n5}{1/n6} = \boxed{1 + \frac{n6}{n5}} \quad (c)$$

$$d = \frac{Y_{n5}(\$)}{Y_{n6}(\$)} = \frac{1/n5}{1/n6} = \boxed{\frac{n6}{n5}} \quad (d)$$

• Show buses:

$$a.c = \frac{\$ c_1 \cdot n_1 \cdot n_3}{n_4 + n_3} \cdot \left(1 + \frac{n6}{n5}\right) = \frac{\$ c_1 \cdot n_1 \cdot n_3}{(n_4 + n_3)} \cdot \frac{(n5 + n6)}{n5} = \frac{\$ c_1 \cdot n_1 \cdot n_3 (n5 + n6)}{n5 (n_4 + n_3)}$$

$$b.c = \frac{n_4}{\$ c_2 \cdot n_2 (n_4 + n_3)} \cdot \left(1 + \frac{n6}{n5}\right) = \frac{n_4}{\$ c_2 \cdot n_2 (n_4 + n_3)} \cdot \frac{(n5 + n6)}{n5} = \frac{n_4 (n5 + n6)}{\$ c_2 \cdot n_2 \cdot n5 (n_4 + n_3)}$$

• Netamonds Φ

$$\frac{V_1(\$)}{V_{n1}(\$)} = - \frac{n6/n5}{1 + \frac{\$ c_1 \cdot n_1 \cdot n_3 (n5 + n6)}{n5 (n_4 + n_3)} + \frac{n_4 (n5 + n6)}{\$ c_2 \cdot n_2 \cdot n5 (n_4 + n_3)}} \quad \frac{\$}{\$}$$

$$\frac{V_1(\$)}{V_{n1}(\$)} = - \frac{[\$ n6/n5]}{\left[\frac{\$^2 \cdot c_1 \cdot n_1 \cdot n_3 (n5 + n6)}{n5 (n_4 + n_3)} + \$ + \frac{n_4 (n5 + n6)}{c_2 \cdot n_2 \cdot n5 (n_4 + n_3)} \right] \cdot \frac{(n5 (n_4 + n_3))}{(c_1 \cdot n_1 \cdot n_3 (n5 + n6))}}$$

$$\boxed{\frac{V_1(\$)}{V_{n1}(\$)} = - \frac{\$ \frac{n6(n_4 + n_3)}{c_1 \cdot n_1 \cdot n_3 (n5 + n6)}}{\$^2 + \$ \cdot \frac{n5 \cdot (n_4 + n_3)}{(c_1 \cdot n_1 \cdot n_3 (n5 + n6))} + \frac{n_4}{c_2 \cdot n_2 \cdot c_1 \cdot n_1 \cdot n_3}} \rightarrow n_{t2c})}$$

• Correlación $\frac{V_1(s)}{V_{in}(s)}$ solo resto división para la simplificación

excepto el signo "-" para luego multiplicar por $\frac{h_2(s)}{h_1(s)}$

luego $\frac{V_2(s)}{V_{in}(s)}$, ya multiplicar por $\frac{V_1(s)}{V_{in}(s)}$ para obtener $\frac{V_2(s)}{V_{in}(s)}$.

$$\frac{V_2(s)}{V_{in}(s)} = \frac{\$ \frac{n_6(n_4+n_3)}{c_1 n_1 n_3 (n_5+n_6)}}{\$^2 + \$ \frac{n_5(n_4+n_3)}{c_1 n_1 n_3 (n_5+n_6)} + \frac{n_4}{c_2 n_2 c_1 n_1 n_3}} \cdot \frac{1/n_2}{\$ \cdot c_2}$$

$$\frac{V_2(s)}{V_{in}(s)} = \frac{\$ \cdot \frac{n_6(n_4+n_3)}{n_2 \cdot c_1 n_1 n_3 (n_5+n_6)}}{\$^3 + \$^2 \cdot \frac{c_2 \cdot n_5}{c_1 n_1 n_3} \frac{(n_4+n_3)}{(n_5+n_6)} + \$ \cdot \frac{n_4}{n_2 \cdot c_1 n_1 n_3}} \cdot \frac{1/c_2}{1/c_2}$$

$$\frac{V_2(s)}{V_{in}(s)} = \frac{\$ \cdot \frac{n_6}{n_2 \cdot c_2 \cdot c_1 n_1 n_3} \frac{(n_4+n_3)}{(n_5+n_6)}}{\$^3 + \$^2 \cdot \frac{n_5}{c_1 n_1 n_3} \frac{(n_4+n_3)}{(n_5+n_6)} + \$ \cdot \frac{n_4}{n_2 \cdot c_2 \cdot c_1 n_1 n_3}}$$

(b. n to b)

$$\frac{V_3(\$)}{Var(\$)} = \frac{\$ \frac{n_6(n_4+n_3)}{c_1 n_1 n_3 (n_5+n_6)}}{\$^2 + \$ \frac{n_5}{c_1 n_1 n_3} \frac{(n_4+n_3)}{(n_5+n_6)} + \frac{n_4}{c_2 n_2 c_1 n_1 n_3}} \cdot \frac{\$ c_1}{1/n_1}$$

$$\frac{V_3(\$)}{Var(\$)} = \frac{\$ \frac{n_6}{c_1 n_1 n_3} \frac{(n_4+n_3)}{(n_5+n_6)} \cdot \$ c_1 n_1}{\$^2 + \$ \frac{n_5}{c_1 n_1 n_3} \frac{(n_4+n_3)}{(n_5+n_6)} + \frac{n_4}{c_2 n_2 c_1 n_1 n_3}}$$

$$\frac{V_3(\$)}{Var(\$)} = \frac{\$^2 \cdot \frac{n_6}{n_3} \frac{(n_4+n_3)}{(n_5+n_6)}}{\$^2 + \$ \frac{n_5}{c_1 n_1 n_3} \frac{(n_4+n_3)}{(n_5+n_6)} + \frac{n_4}{c_2 n_2 c_1 n_1 n_3}}$$

b n t o a)